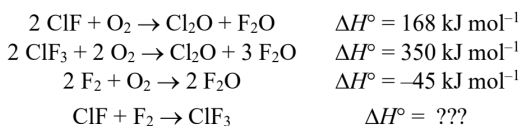


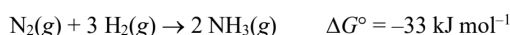
Thermodynamics

1. What is the standard enthalpy change for the reaction of ClF with F₂ to form ClF₃? (All species are in the gas phase.)



- (A) -46 kJ mol⁻¹ (B) -69 kJ mol⁻¹
(C) -114 kJ mol⁻¹ (D) -136 kJ mol⁻¹

2. At 25 °C, the standard free energy change for the gas-phase reaction of hydrogen and nitrogen to form NH₃ is -33.0 kJ mol⁻¹.



Under what circumstances is $\Delta G > 0$ at 298 K?

- (A) Partial pressures of N₂ and H₂ = 0.1 bar, partial pressure of NH₃ = 10 bar
(B) Partial pressures of N₂ and H₂ = 10 bar, partial pressure of NH₃ = 0.1 bar
(C) All partial pressures = 1×10^{-2} bar
(D) Under no possible circumstances

3. In a coal-fired power plant, coal is burned to generate heat that boils water. 2.5 kg of coal is burned to heat 500.0 kg of water at 95 °C, ultimately converting 30.0 kg to steam. Which type of coal was burned? (For liquid water near 100 °C, $C_p = 4.21 \text{ kJ kg}^{-1} \text{ K}^{-1}$ and $\Delta H^\circ_{\text{vap}} = 2260 \text{ kJ kg}^{-1}$.)

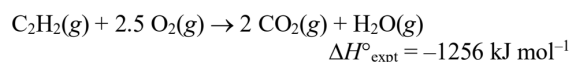
- (A) Lignite, $\Delta H^\circ_{\text{combustion}} = 15 - 20 \text{ kJ g}^{-1}$
(B) Sub-bituminous, $\Delta H^\circ_{\text{combustion}} = 23 - 28 \text{ kJ g}^{-1}$
(C) Bituminous, $\Delta H^\circ_{\text{combustion}} = 26 - 30 \text{ kJ g}^{-1}$
(D) Anthracite, $\Delta H^\circ_{\text{combustion}} = 30 - 33 \text{ kJ g}^{-1}$

4. In a vacuum calorimetry experiment, high-energy plasma discharges between Ti(s) electrodes are used to electrolyze water in dilute KOH(aq). During an experiment, the electrical input is 500. kJ and the calorimeter records 610. kJ of heat release into the surroundings. Which best explains this experimental result?

- (A) The electrodes undergo partial oxidation, an additional reaction not accounted for in electrolysis.
(B) The system's large specific heat capacity increases the total heat registered per unit of electrical energy.
(C) The overall process is endothermic, so the surroundings must release heat to maintain equilibrium.
(D) The entropy increase from gas formation causes the system to do work, resulting in additional temperature rise.

5. The standard enthalpy of combustion of ethyne can be estimated from the tabulated bond dissociation enthalpies (BDE) shown below to be -982 kJ mol⁻¹. The experimental $\Delta H^\circ_{\text{combustion}}$ is -1256 kJ mol⁻¹. Which is the best explanation for the discrepancy?

Bond	BDE (kJ mol ⁻¹)
C≡C	839
C-H	412
O=O	498
C=O	741
O-H	463



- (A) The tabulated C-H BDE does not account for the unusual strength of the bond formed to the *sp*-hybridized carbon in C₂H₂.
(B) The tabulated O=O BDE does not account for the unusual weakness of the bond in dioxygen due to its paramagnetism.
(C) The tabulated C=O BDE does not account for the stabilization due to electronic delocalization in carbon dioxide.
(D) The tabulated O-H BDE does not account for the stabilization due to hydrogen bonding in water.

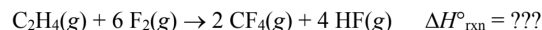
6. For which reaction is $\Delta S^\circ > 0$?

- (A) $\text{P}_4(\text{s}) \rightarrow 4 \text{P}(\text{s}, \text{red})$
(B) $\text{CO}_2(\text{s}) \rightarrow \text{CO}_2(\text{g})$
(C) $3 \text{O}_2(\text{g}) \rightarrow 2 \text{O}_3(\text{g})$
(D) $2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2 \text{SO}_3(\text{g})$

7. The standard enthalpy of fusion of 1,4-dichlorobenzene is 18.2 kJ mol⁻¹ and its standard entropy of fusion is 55.3 J mol⁻¹ K⁻¹. What is the melting point of 1,4-dichlorobenzene?

- (A) 37.1 °C (B) 56.0 °C
(C) 101 °C (D) 329 °C

8. What is $\Delta H^\circ_{\text{rxn}}$ for the gas-phase reaction of ethene and fluorine shown below?

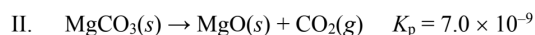
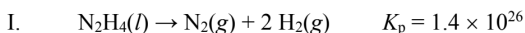


Substance	ΔH°_f , kJ mol ⁻¹
C ₂ H ₄ (g)	52.4
CF ₄ (g)	-679.0
HF(g)	-273.3

- (A) -2503.6 kJ mol⁻¹ (B) -1004.7 kJ mol⁻¹
(C) 1004.7 kJ mol⁻¹ (D) 2503.6 kJ mol⁻¹

Thermodynamics

9. Which reactions are spontaneous under standard conditions?



- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

10. Solid gallium melts at 29.8 °C with an enthalpy of fusion of 5.59 kJ mol⁻¹. The heat capacity of solid gallium is 26.1 J mol⁻¹ K⁻¹ and the heat capacity of liquid gallium is 28.5 J mol⁻¹ K⁻¹. If equal masses of solid gallium at 20.0 °C and liquid gallium at 60.0 °C are mixed in a well-insulated container, what is the temperature after the mixture has attained equilibrium?

- (A) 29.8 °C (B) 40.0 °C
(C) 40.4 °C (D) 40.9 °C

11. What is the K_{sp} of AgBr at 298 K?

Species	ΔG°_f (298 K), kJ mol ⁻¹
$\text{Ag}^+(aq)$	77.1
$\text{Br}^-(aq)$	-102.8
$\text{AgBr}(s)$	-96.9

- (A) 3.3×10^{-13} (B) 3.1×10^{-14}
(C) 1.0×10^{-17} (D) 3.2×10^{-22}

12. For a chemical reaction under conditions where it is at equilibrium, which statements must be true?

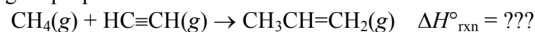
- I. $\Delta S_{\text{rxn}} > 0$ II. $\Delta G_{\text{rxn}} = 0$

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

13. Which statement about phase transitions is correct?

- (A) Melting is endothermic and freezing is exothermic.
(B) Vaporization is exothermic and condensation is endothermic.
(C) The enthalpy change for a spontaneous phase transition is always positive.
(D) The entropy change for a spontaneous phase transition is always positive.

14. Based on the given bond dissociation enthalpies (BDEs), what is $\Delta H^\circ_{\text{rxn}}$ for the addition of methane to ethyne to give propene?



Bond	BDE, kJ mol ⁻¹	Bond	BDE, kJ mol ⁻¹
C-H	415	C=C	611
C-C	345	C≡C	837

- (A) -119 kJ mol⁻¹ (B) -196 kJ mol⁻¹
(C) -259 kJ mol⁻¹ (D) -296 kJ mol⁻¹

15. Which statements must be true for a nonspontaneous gas-phase chemical reaction at constant temperature and pressure?

- I. $\Delta G > 0$ II. $K_p < 1$

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

16. Benzene, C₆H₆, has $\Delta H^\circ_{\text{vap}} = 30.7 \text{ kJ mol}^{-1}$ at its normal boiling point of 80.1 °C. What is $\Delta S^\circ_{\text{vap}}$ at this temperature?

- (A) 10.5 J mol⁻¹ K⁻¹ (B) 86.9 J mol⁻¹ K⁻¹
(C) 255 J mol⁻¹ K⁻¹ (D) 383 J mol⁻¹ K⁻¹

17. Which reaction has $\Delta S^\circ < 0$?

- (A) $\text{C}(s, \text{graphite}) \rightarrow \text{C}(s, \text{diamond})$
(B) $\text{N}_2\text{O}_4(g) \rightarrow 2 \text{NO}_2(g)$
(C) $\text{I}_2(s) \rightarrow \text{I}_2(g)$
(D) $\text{NH}_4^+(aq) + \text{CH}_3\text{COO}^-(aq) \rightarrow \text{NH}_3(aq) + \text{CH}_3\text{COOH}(aq)$

18. What is the K_{sp} of PbCl₂ at 298 K?

Species	ΔH°_f , kJ mol ⁻¹	S° , J mol ⁻¹ K ⁻¹
$\text{Pb}^{2+}(aq)$	-1.7	10.5
$\text{PbCl}_2(s)$	-359.4	136.0
$\text{Cl}^-(aq)$	-167.2	56.5

- (A) 1.2×10^{-11} (B) 1.8×10^{-5}
(C) 8.2×10^{-5} (D) 3.7×10^{-4}

Thermodynamics

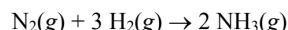
19. Given the standard enthalpies of the gas-phase reactions of hydrogen sulfide, what is the standard enthalpy of reaction for the gas-phase combustion of carbon disulfide?

Reaction	$\Delta H^\circ_{\text{rxn}}, \text{kJ mol}^{-1}$
$\text{H}_2\text{S} + 1.5 \text{O}_2 \rightarrow \text{SO}_2 + \text{H}_2\text{O}$	-518.2
$2 \text{H}_2\text{S} + \text{CO}_2 \rightarrow \text{CS}_2 + 2 \text{H}_2\text{O}$	67.8
$\text{CS}_2 + 3 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{SO}_2$	???

(A) $-450.4 \text{ kJ mol}^{-1}$ (B) $-586.0 \text{ kJ mol}^{-1}$
 (C) $-968.6 \text{ kJ mol}^{-1}$ (D) $-1104.2 \text{ kJ mol}^{-1}$

20. Which substance has a standard enthalpy of formation, ΔH°_f , that is greater than zero?
- (A) $\text{O}_2(\text{g})$ (B) $\text{F}_2(\text{g})$ (C) $\text{S}_2(\text{g})$ (D) $\text{Cl}_2(\text{g})$

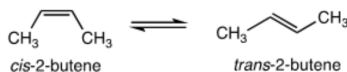
21. For the reaction



$\Delta G_{\text{rxn}} = -33 \text{ kJ mol}^{-1}$ when all gases have a partial pressure of 1.0 bar. Changing the pressure of which gas to 2.0 bar will give the most negative value of ΔG_{rxn} ?

- (A) N_2
 (B) H_2
 (C) NH_3
 (D) The value of ΔG_{rxn} will be the same regardless of the gas pressures.
22. Gallium metal melts at 29.8°C with $\Delta H^\circ_{\text{fusion}} = 5.59 \text{ kJ mol}^{-1}$. If equal masses of solid gallium initially at 20.0°C and liquid gallium initially at 35.0°C are mixed in a well-insulated container, what percentage of the gallium will be solid after the mixture has reached equilibrium? The molar heat capacity of solid gallium is $26.2 \text{ J mol}^{-1} \text{ K}^{-1}$ and the molar heat capacity of liquid gallium is $28.0 \text{ J mol}^{-1} \text{ K}^{-1}$.
- (A) 34.7% (B) 49.0% (C) 51.0% (D) 65.3%

23. *cis*-2-butene isomerizes to *trans*-2-butene with $\Delta G^\circ_{\text{rxn}} = -3.0 \text{ kJ mol}^{-1}$ at 298 K.

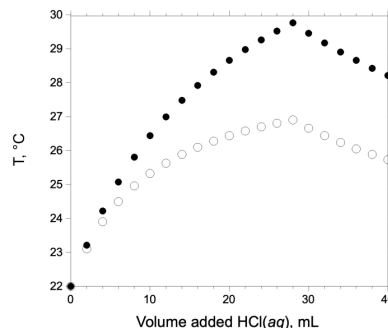


$$\Delta G^\circ_{\text{rxn}} = -3.0 \text{ kJ mol}^{-1}$$

Some *cis*-2-butene is introduced into a vessel maintained at 298 K, along with a catalyst for isomerization. After equilibrium is attained, what is the mole fraction of *cis*-2-butene in the vessel?

- (A) 0.00 (B) 0.23 (C) 0.30 (D) 0.50

24. A certain volume of $\text{NaOH}(\text{aq})$ is titrated against $\text{HCl}(\text{aq})$ in a well-insulated vessel with constant stirring. Each solution is initially at 22.0°C , and the temperature of the reaction mixture is monitored after each 2.0 mL of hydrochloric acid is added.



The data in closed circles correspond to an experiment conducted with 20.0 mL NaOH solution and 1.0 M HCl . The open circles were carried out using the same concentration but a different volume of NaOH solution, and using a different concentration of aqueous HCl . Which volume of NaOH and concentration of HCl correspond to the conditions that give the open circles?

	Volume NaOH	Concentration HCl
Closed circles:	20 mL	1.0 M
Open circles:		
(A)	40 mL	2.0 M
(B)	30 mL	2.0 M
(C)	15 mL	0.5 M
(D)	10 mL	0.5 M

25. The amount of energy released by a chemical process is always equal to that absorbed by the system and surroundings. This is a consequence of which law?
- (A) The Zeroth Law of Thermodynamics
 (B) The First Law of Thermodynamics
 (C) The Second Law of Thermodynamics
 (D) The Third Law of Thermodynamics

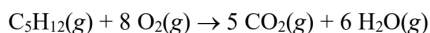
26. What is ΔG°_f of $\text{NO}(\text{g})$ at 298 K?

Substance	$\Delta H^\circ_f, \text{kJ mol}^{-1}$	$S^\circ, \text{J mol}^{-1} \text{ K}^{-1}$
$\text{N}_2(\text{g})$	0	191.6
$\text{O}_2(\text{g})$	0	205.2
$\text{NO}(\text{g})$	90.3	210.8

- (A) 27.5 kJ mol^{-1} (B) 86.6 kJ mol^{-1}
 (C) $145.7 \text{ kJ mol}^{-1}$ (D) $153.1 \text{ kJ mol}^{-1}$

Thermodynamics

27. Which statements about entropy, enthalpy, and energy are correct for the reaction below?



I. $\Delta S^\circ > 0$ II. $\Delta H^\circ > \Delta E^\circ$

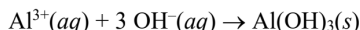
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

28. 4.00 mL of 2.00 M aqueous NaOH is added to 100.0 mL of 0.200 M aqueous NaHSO₄, all initially at 25.00 °C, in a well-insulated flask. Neglecting the effects of dilution, and assuming that volumes are additive and that all solutions have the same density and specific heat capacity as pure water, what is the final temperature of the mixture?

Species	ΔH°_f , kJ mol ⁻¹
OH ⁻ (aq)	-230
HSO ₄ ⁻ (aq)	-886
SO ₄ ²⁻ (aq)	-909
H ₂ O(l)	-286

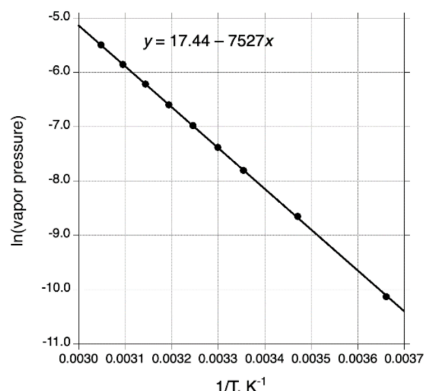
- (A) 25.18 °C (B) 26.06 °C
(C) 26.45 °C (D) 28.63 °C

29. The K_{sp} of Al(OH)₃ is 2.0×10^{-31} at 298 K. What is ΔG° (at 298 K) for the precipitation of Al(OH)₃ according to the equation below?



- (A) -175 kJ mol⁻¹ (B) 14.7 kJ mol⁻¹
(C) 70.6 kJ mol⁻¹ (D) 175 kJ mol⁻¹

30. The vapor pressure of solid I₂ is measured (in bar) over the temperature range 0–55 °C. The natural logarithms of these vapor pressures are plotted as a function of the reciprocal of the absolute temperature as shown below. What is the standard enthalpy of sublimation ($\Delta H^\circ_{\text{subl}}$) of I₂(s)?

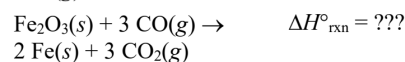


- (A) 7.53 kJ mol⁻¹ (B) 17.4 kJ mol⁻¹
(C) 19.4 kJ mol⁻¹ (D) 62.6 kJ mol⁻¹

31. A container with 100.0 g of ice at 0 °C is placed in a humid room whose temperature is 40 °C. The ice melts as water vapor condenses into the container. Assuming that all the heat transferred to the container comes from condensation, how much water will have condensed in the container once all the ice is melted and has reached 40 °C? The heat of fusion of ice is 334 J g⁻¹ and the heat of vaporization of water is 2260 J g⁻¹.

- (A) 13.8 g (B) 14.8 g (C) 22.2 g (D) 677 g

32. The standard enthalpy of formation of hematite, Fe₂O₃(s), is -825.5 kJ mol⁻¹. What is the standard enthalpy of reaction for the reduction of hematite with carbon monoxide to give metallic iron as shown below?



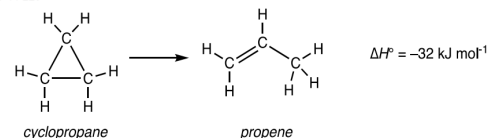
- (A) -23.5 kJ mol⁻¹ (B) -542.5 kJ mol⁻¹
(C) -849.0 kJ mol⁻¹ (D) -1674.5 kJ mol⁻¹

33. Which of the following equations correctly describe the change in internal energy, ΔE , of a reaction taking place in a bomb calorimeter?

I. $\Delta E = q$
II. $\Delta E = \Delta H - V\Delta P$

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

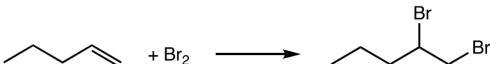
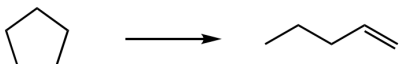
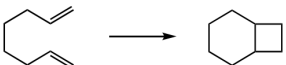
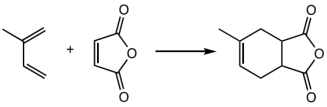
34. Cyclopropane isomerizes exothermically to propene as shown.



The C–C single bond in propene has an estimated bond dissociation enthalpy (BDE) of 345 kJ mol⁻¹, while the C=C double bond in propene has a BDE of 611 kJ mol⁻¹. Assuming that the C–H bonds in propene and cyclopropane all have the same BDE, what is the estimated BDE of a C–C single bond in cyclopropane?

- (A) 308 kJ mol⁻¹ (B) 329 kJ mol⁻¹
(C) 423 kJ mol⁻¹ (D) 924 kJ mol⁻¹

Thermodynamics

35. Two reactions have similar $\Delta H^\circ_{\text{rxn}}$ values ($\Delta H^\circ_{\text{rxn}(1)} \approx \Delta H^\circ_{\text{rxn}(2)}$), but reaction (1) has a much smaller standard entropy change than reaction (2) ($\Delta S^\circ_{\text{rxn}(1)} \ll \Delta S^\circ_{\text{rxn}(2)}$). At 298 K, which statements about these two reactions must be correct?
- Reaction (1) must have a larger equilibrium constant ($K_{\text{eq}(1)} > K_{\text{eq}(2)}$).
 - Reaction (1) must have a larger Arrhenius prefactor ($A_{(1)} > A_{(2)}$).
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
36. Which reaction has $\Delta S^\circ_{\text{rxn}} > 0$? All reactants and products are liquids.
- (A) 
- (B) 
- (C) 
- (D) 
37. Burning 48.0 g of graphite in excess oxygen under standard conditions releases 1574.0 kJ of heat. What is ΔH°_f of $\text{CO}_2(\text{g})$?
- (A) 1574 kJ mol⁻¹ (B) -1574 kJ mol⁻¹
(C) -394 kJ mol⁻¹ (D) 32.8 kJ mol⁻¹
38. Two metal samples, labeled A and B, absorb the same amount of heat. Sample A has a mass of 10.0 g, and its temperature increases by 38 °C. Sample B has a mass of 20.0 g, and its temperature increases by 23 °C. Which sample has the greater specific heat capacity?
- (A) Sample A
(B) Sample B
(C) Both samples have the same specific heat capacity.
(D) It is impossible to determine from the information given.
39. A reaction has $\Delta S^\circ > 0$ and $\Delta H^\circ > 0$. Which statement about this reaction must be correct?
- (A) If carried out in a well-insulated flask, the temperature of the reaction mixture will decrease.
(B) It will occur spontaneously at 298 K and 1 atm pressure.
(C) As the temperature is raised, K_{eq} for this reaction decreases.
(D) The reaction has more moles of products than it has moles of reactants.
40. What is the boiling point of water in a pressure cooker with a pressure of 2.00 atm? (The enthalpy of vaporization of water is 40.7 kJ mol⁻¹.)
- (A) 101 °C (B) 121 °C (C) 141 °C (D) 200 °C
41. The gas-phase bromination of propene has a standard enthalpy of reaction of -122.5 kJ mol⁻¹.
- $$\text{C}_3\text{H}_6(\text{g}) + \text{Br}_2(\text{g}) \rightarrow \text{C}_3\text{H}_5\text{Br}_2(\text{g})$$
- $$\Delta H^\circ = -122.5 \text{ kJ mol}^{-1}$$
- | Substance | ΔH°_f , kJ mol ⁻¹ |
|----------------------------------|---|
| $\text{C}_3\text{H}_6(\text{g})$ | 20.4 |
| $\text{Br}_2(\text{g})$ | 30.9 |
- What is the standard enthalpy of formation of $\text{C}_3\text{H}_5\text{Br}_2(\text{g})$?
- (A) -71.2 kJ mol⁻¹ (B) -102.1 kJ mol⁻¹
(C) -142.9 kJ mol⁻¹ (D) -173.8 kJ mol⁻¹
42. Titanium has a normal melting point of 1668 °C and a molar enthalpy of fusion of 14.15 kJ mol⁻¹. The standard molar entropy of liquid titanium is 97.53 J mol⁻¹ K⁻¹ at 1668 °C. What is the standard molar entropy of solid titanium at this temperature?
- (A) 89.05 J mol⁻¹ K⁻¹ (B) 90.24 J mol⁻¹ K⁻¹
(C) 97.52 J mol⁻¹ K⁻¹ (D) 104.82 J mol⁻¹ K⁻¹
43. For which reaction is $\Delta H^\circ_{\text{rxn}}$ equal to ΔH°_f of $\text{CaSO}_3(\text{s})$?
- (A) $\text{Ca}(\text{s}) + \frac{1}{8} \text{S}_8(\text{s}) + \frac{3}{2} \text{O}_2(\text{g}) \rightarrow \text{CaSO}_3(\text{s})$
(B) $8 \text{Ca}(\text{s}) + \text{S}_8(\text{s}) + 12 \text{O}_2(\text{g}) \rightarrow 8 \text{CaSO}_3(\text{s})$
(C) $8 \text{Ca}(\text{s}) + \text{S}_8(\text{g}) + 12 \text{O}_2(\text{g}) \rightarrow 8 \text{CaSO}_3(\text{s})$
(D) $\text{CaO}(\text{s}) + \text{SO}_2(\text{g}) \rightarrow \text{CaSO}_3(\text{s})$
44. A 2.00 g sample of ice at 0.0 °C is placed in 50.0 g of water initially at 25.0 °C in an insulated container. What is the final temperature after the system has achieved equilibrium? (For ice, $\Delta H^\circ_{\text{fusion}} = 6.01 \text{ kJ mol}^{-1}$.)
- (A) 21.0 °C (B) 21.8 °C
(C) 22.5 °C (D) 24.0 °C

Thermodynamics

45. Ethanol has a normal boiling point of 78.3 °C and a standard heat of vaporization ($\Delta H^\circ_{\text{vap}}$) of 38.6 kJ mol⁻¹. What is the vapor pressure of ethanol at 45.0 °C?

(A) 0.25 atm (B) 0.57 atm
(C) 0.87 atm (D) 0.91 atm

46. If the average carbon-hydrogen bond dissociation enthalpy in ethane is 416 kJ mol⁻¹, what is the bond dissociation enthalpy of the carbon-carbon bond in ethane?

Species	ΔH°_f , kJ mol ⁻¹
C ₂ H ₆ (g)	-84.7
H(g)	217.9
C(g)	718.4

(A) 164 kJ mol⁻¹ (B) 333 kJ mol⁻¹
(C) 386 kJ mol⁻¹ (D) 404 kJ mol⁻¹

47. A reaction has $K_{\text{eq}} = 20$ at 298 K, and K_{eq} increases with increasing temperature between 298 K and 350 K. What may be concluded from these observations?

I. $\Delta G^\circ_{\text{rxn}} < 0$ at 330 K
II. $\Delta S^\circ_{\text{rxn}} > 0$ at 330 K

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

48. A sample of water in a movable piston is maintained at 100.0 °C with an external pressure p . Which statement best describes the transformation of some of the liquid water into water vapor?

(A) It is spontaneous if $p \geq 1$ atm.
(B) It is reversible if $p = 1$ atm and spontaneous if $p > 1$ atm.
(C) It is nonspontaneous if $p \geq 1$ atm.
(D) It is reversible if $p = 1$ atm and nonspontaneous if $p > 1$ atm.

49. The specific heat capacity of iron is 0.461 J g⁻¹ K⁻¹ and that of titanium is 0.544 J g⁻¹ K⁻¹. A sample consisting of a mixture of 10.0 g Fe and 10.0 g Ti at 100.0 °C loses 200. J of heat to the environment. What is the final temperature of the sample?

(A) 89.9 °C (B) 80.1 °C
(C) 60.2 °C (D) 39.8 °C

50. Which sample has the greatest entropy at a given temperature?

(A) 2 mol HBr(s)
(B) 2 mol HBr(l)
(C) 2 mol HBr(g)
(D) 1 mol H₂(g) + 1 mol Br₂(l)

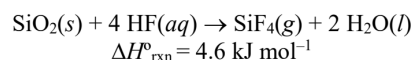
51. For an endothermic reaction to be spontaneous under standard conditions at constant pressure at some temperature T , which must be true?

(A) The entropy change ΔS° must be positive and greater than $\Delta H^\circ/T$.
(B) The entropy change ΔS° must be positive and less than $\Delta H^\circ/T$.
(C) The entropy change ΔS° must be negative and greater than $-\Delta H^\circ/T$.
(D) The entropy change ΔS° must be negative and less than $-\Delta H^\circ/T$.

52. For a constant-pressure process, what is the difference between the internal energy change (ΔU) and the enthalpy change (ΔH)?

(A) Heat (B) Work
(C) Entropy (D) Gibbs free energy

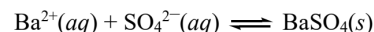
53. The standard enthalpy of reaction for the dissolution of silica in aqueous HF is 4.6 kJ mol⁻¹. What is the standard enthalpy of formation of SiF₄(g)?



Compound	SiO ₂ (s)	HF(aq)	H ₂ O(l)	SiF ₄ (g)
ΔH°_f , kJ mol ⁻¹	-910.9	-320.1	-285.8	???

(A) -1624.3 kJ mol⁻¹ (B) -1615.1 kJ mol⁻¹
(C) -949.8 kJ mol⁻¹ (D) -940.6 kJ mol⁻¹

54. The K_{sp} of BaSO₄ at 298 K is 1.1×10^{-10} . What is ΔG° at 298 K for the following reaction?



(A) -57 kJ mol⁻¹ (B) -25 kJ mol⁻¹
(C) 25 kJ mol⁻¹ (D) 57 kJ mol⁻¹

Thermodynamics

55. To 100.0 g water at 25.00 °C in a well-insulated container is added a block of aluminum initially at 100.0 °C. The temperature of the water once the system reaches thermal equilibrium is 28.00 °C. What is the mass of the aluminum block? (The specific heat capacity of Al is 0.900 J g⁻¹ K⁻¹.)

(A) 4.17 g (B) 18.6 g (C) 19.4 g (D) 130. g

56. The standard enthalpy of formation, ΔH_f° , of the compound MgO(s) is equal to the standard enthalpy change for which reaction?

(A) $\text{Mg}(s) + \frac{1}{2} \text{O}_2(g) \rightarrow \text{MgO}(s)$
 (B) $2 \text{Mg}(s) + \text{O}_2(g) \rightarrow 2 \text{MgO}(s)$
 (C) $\text{Mg}(g) + \text{O}(g) \rightarrow \text{MgO}(s)$
 (D) $\text{Mg}^{2+}(aq) + \text{O}^{2-}(aq) \rightarrow \text{MgO}(s)$

57. The enthalpy of formation of XeF₂(g) is -108 kJ mol⁻¹ and the bond dissociation enthalpy of the F-F bond is 155 kJ mol⁻¹. What is the average bond dissociation enthalpy of a Xe-F bond?

(A) 47 kJ mol⁻¹ (B) 54 kJ mol⁻¹
 (C) 132 kJ mol⁻¹ (D) 263 kJ mol⁻¹

58. What is the standard Gibbs free energy of formation, ΔG_f° , of NH₃(g) at 298 K?

Substance	ΔH_f° , kJ mol ⁻¹	S° , J mol ⁻¹ K ⁻¹
H ₂ (g)	0	131
N ₂ (g)	0	192
NH ₃ (g)	-46	193

(A) -104 kJ mol⁻¹ (B) -16 kJ mol⁻¹
 (C) -7 kJ mol⁻¹ (D) 13 kJ mol⁻¹

59. The vaporization of a liquid at a certain temperature and pressure is spontaneous. For this process, which of the inequalities regarding the Gibbs free energy G and the internal energy E are correct?

I. $\Delta G < 0$ II. $\Delta E < \Delta H$
 (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

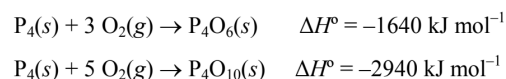
60. Which of the following reactions takes place with an increase in entropy under standard conditions?

(A) $\text{NH}_4^+(aq) + \text{CH}_3\text{COO}^-(aq) \rightarrow \text{NH}_3(aq) + \text{CH}_3\text{COOH}(aq)$
 (B) $\text{CaO}(s) + \text{CO}_2(g) \rightarrow \text{CaCO}_3(s)$
 (C) $\text{NH}_3(g) + \text{HCl}(g) \rightarrow \text{NH}_4^+(aq) + \text{Cl}^-(aq)$
 (D) $\text{C}_2\text{H}_4(g) + \text{Br}_2(l) \rightarrow \text{C}_2\text{H}_4\text{Br}_2(l)$

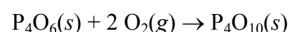
61. A sample of 54.0 g of methanol is heated from 25.0 °C to 35.0 °C. How much heat is required? The specific heat capacity of methanol is 2.48 J g⁻¹ K⁻¹.

(A) 0.00459 J (B) 0.0747 J
 (C) 1340 J (D) 4690 J

62. Given the standard enthalpy changes for the reactions:

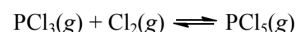


Calculate the standard enthalpy change ΔH° for the following reaction:



(A) $-4.58 \times 10^3 \text{ kJ mol}^{-1}$ (B) $-1.30 \times 10^3 \text{ kJ mol}^{-1}$
 (C) 1.79 kJ mol⁻¹ (D) $4.82 \times 10^6 \text{ kJ mol}^{-1}$

63. At what temperature is the following reaction at equilibrium when all substances are at standard pressure? Assume that entropies and enthalpies of reaction do not vary with temperature.



Substance	ΔH_f° , kJ mol ⁻¹	S° , J mol ⁻¹ K ⁻¹
PCl ₃ (g)	-288.7	311.6
Cl ₂ (g)	0	223.1
PCl ₅ (g)	-374.9	364.2

(A) 506 K (B) 1640 K
 (C) 1980 K (D) 4260 K

64. Given the bond dissociation enthalpies (BDE) below, what is the approximate ΔH_f° for H₂O(g)?

Bond	BDE, kJ mol ⁻¹	Bond	BDE, kJ mol ⁻¹
H-H	432	O-O	146
O-H	467	O=O	495

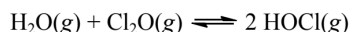
(A) -934 kJ mol⁻¹ (B) -510. kJ mol⁻¹
 (C) -429 kJ mol⁻¹ (D) -255 kJ mol⁻¹

65. Which of the following are true for a spontaneous process in a system at constant temperature and pressure?

I. $\Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$ II. $\Delta G_{\text{sys}} < 0$
 (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

Thermodynamics

66. What is ΔG°_f of $\text{HOCl}(g)$ at 298 K?



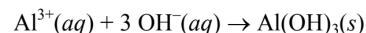
$$K_{\text{eq}}(298 \text{ K}) = 0.089$$

- (A) -6.0 kJ mol^{-1}
 (B) -3.0 kJ mol^{-1}
 (C) 3.0 kJ mol^{-1}
 (D) It cannot be determined from the information given.
67. The standard enthalpy of formation, ΔH°_f , for $\text{HCOOH}(l)$ is equal to the standard enthalpy change for which reaction?
- (A) $\text{C}(g) + 2 \text{H}(g) + 2 \text{O}(g) \rightarrow \text{HCOOH}(l)$
 (B) $\text{C}(s) + \text{H}_2(g) + \text{O}_2(g) \rightarrow \text{HCOOH}(l)$
 (C) $\text{C}(g) + \text{H}_2(g) + \text{O}_2(g) \rightarrow \text{HCOOH}(l)$
 (D) $\text{CO}_2(g) + \text{H}_2(g) \rightarrow \text{HCOOH}(l)$
68. 40.0 mL of 0.200 M aqueous NaOH is added to 200.0 mL of 0.100 M aqueous NaHCO_3 in a flask maintained at 25 °C. Neglecting the effects of dilution, what is q for this reaction?

	$\Delta H^\circ_f, \text{ kJ mol}^{-1}$
$\text{OH}^-(aq)$	-230
$\text{HCO}_3^-(aq)$	-692
$\text{CO}_3^{2-}(aq)$	-677
$\text{H}_2\text{O}(l)$	-286

- (A) -41 J (B) -74 J (C) -330 J (D) -820 J
69. In a well-insulated vessel, 50.0 g ice at 0.0 °C is added to 350. g water at 32.0 °C. What is the final temperature when the mixture reaches equilibrium? (The heat of fusion of ice is 334 J g^{-1} .)
- (A) 18.0 °C (B) 20.6 °C
 (C) 22.0 °C (D) 28.0 °C
70. Which is the best explanation for the negative sign of ΔS° in the following reaction?
- $$\text{CaSO}_4(s) \rightarrow \text{Ca}^{2+}(aq) + \text{SO}_4^{2-}(aq)$$
- $$\Delta S^\circ = -143 \text{ J mol}^{-1} \text{ K}^{-1}$$
- (A) There are more ways of arranging the Ca^{2+} and SO_4^{2-} ions in aqueous solution than in the crystal lattice.
 (B) Solid CaSO_4 is a network covalent solid, whereas it separates into ions in aqueous solution.
 (C) Aqueous Ca^{2+} and SO_4^{2-} ions are tightly solvated, decreasing the number of ways of arranging water molecules when the solid dissolves.
 (D) Calcium sulfate dissolves exothermically, leading to a net loss of entropy.

71. The K_{sp} of $\text{Al}(\text{OH})_3$ is 2.0×10^{-31} at 298 K. What is ΔG° (at 298 K) for the precipitation of $\text{Al}(\text{OH})_3$ according to the equation below?

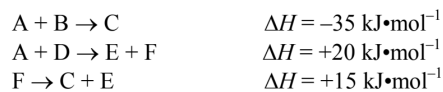


- (A) -175 kJ mol^{-1} (B) 14.7 kJ mol^{-1}
 (C) 70.6 kJ mol^{-1} (D) 175 kJ mol^{-1}
72. The bond dissociation enthalpies of the H-H bond and the H-Cl bond are 435 kJ mol^{-1} and 431 kJ mol^{-1} , respectively. The ΔH°_f of $\text{HCl}(g)$ is -92 kJ mol^{-1} . What is the bond dissociation enthalpy of the Cl-Cl bond?
- (A) 88 kJ mol^{-1} (B) 96 kJ mol^{-1}
 (C) 188 kJ mol^{-1} (D) 243 kJ mol^{-1}
73. Which of the following is a mathematical statement of the first law of thermodynamics?
- (A) $\Delta V = (nR/P)\Delta T$ (B) $\Delta E = q + w$
 (C) $\Delta H = \Delta E + P\Delta V$ (D) $\Delta G = \Delta H - T\Delta S$
74. The enthalpy change under standard conditions for which of the reactions below would be equal to the ΔH°_f of $\text{NaOH}(s)$?
- (A) $\text{Na}(s) + \text{H}_2\text{O}(l) \rightarrow \text{NaOH}(s) + \frac{1}{2} \text{H}_2(g)$
 (B) $\text{Na}(s) + \frac{1}{2} \text{O}_2(g) + \frac{1}{2} \text{H}_2(g) \rightarrow \text{NaOH}(s)$
 (C) $\text{Na}(s) + \frac{1}{2} \text{H}_2\text{O}_2(l) \rightarrow \text{NaOH}(s)$
 (D) $\text{Na}^+(aq) + \text{OH}^-(aq) \rightarrow \text{NaOH}(s)$
75. A 37.5 g piece of gold at 83.0 °C is added to 100. g H_2O at 22.0 °C in a well-insulated cup. What is the temperature after the system comes to equilibrium? (The specific heat capacity of Au is $0.129 \text{ J} \cdot \text{g}^{-1} \cdot \text{K}^{-1}$)
- (A) 22.7 °C (B) 23.0 °C
 (C) 25.0 °C (D) 52.5 °C
76. Which of these reactions has $\Delta S^\circ > 0$?

- (A) $\text{S}_8(l) \rightarrow \text{S}_8(s, \text{monoclinic})$
 (B) $\text{H}_2(g) + \text{O}_2(g) \rightarrow \text{H}_2\text{O}_2(aq)$
 (C) $\text{H}_2(g) + 2 \text{Ag}^+(aq) \rightarrow 2 \text{H}^+(aq) + 2 \text{Ag}(s)$
 (D) $\text{PCl}_5(g) \rightarrow \text{PCl}_3(g) + \text{Cl}_2(g)$

Thermodynamics

77. Given the enthalpy changes:



What is ΔH for the reaction $2\text{A} + \text{B} + \text{D} \rightarrow 2\text{F}$?

- (A) $0 \text{ kJ}\cdot\text{mol}^{-1}$ (B) $-30 \text{ kJ}\cdot\text{mol}^{-1}$
(C) $-40 \text{ kJ}\cdot\text{mol}^{-1}$ (D) $-70 \text{ kJ}\cdot\text{mol}^{-1}$

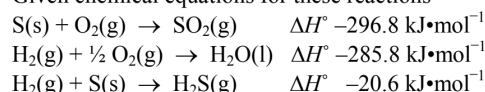
78. The K_a of phosphoric acid, H_3PO_4 , is 7.6×10^{-3} at 25°C . For the reaction



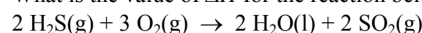
$\Delta H^\circ = -14.2 \text{ kJ/mol}$. What is the K_a of H_3PO_4 at 60°C ?

- (A) 4.2×10^{-3} (B) 6.8×10^{-3}
(C) 8.5×10^{-3} (D) 1.8×10^{-2}

79. Given chemical equations for these reactions

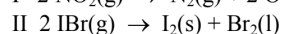
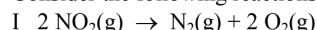


What is the value of ΔH for the reaction below?



- (A) $-603.2 \text{ kJ}\cdot\text{mol}^{-1}$ (B) $-562.0 \text{ kJ}\cdot\text{mol}^{-1}$
(C) $-1206.4 \text{ kJ}\cdot\text{mol}^{-1}$ (D) $-1124.0 \text{ kJ}\cdot\text{mol}^{-1}$

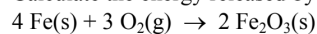
80. Consider the following reactions



For which reaction is $\Delta S^\circ < 0$?

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

81. Calculate the energy released by the reaction



when a 55.8 g sample of iron reacts completely with 1.00 mole of oxygen. The enthalpy of formation (ΔH_f°) of $\text{Fe}_2\text{O}_3(\text{s})$ is $-826 \text{ kJ}\cdot\text{mol}^{-1}$.

- (A) 206 kJ (B) 413 kJ (C) 826 kJ (D) 1650 kJ

82. The specific heat capacities of three metals are given below.

Metal	Fe	Pb	Zn
Specific heat, $\text{J}\cdot\text{g}^{-1}\cdot^\circ\text{C}^{-1}$	0.470	0.130	0.388

If 1.00 g of each metal is heated to 100°C and added to 10.0 g of H_2O at 25.0°C , what is the order of the temperatures of the final mixtures from the lowest to the highest?

- (A) $\text{Fe} < \text{Zn} < \text{Pb}$ (B) $\text{Pb} < \text{Zn} < \text{Fe}$
(C) $\text{Zn} < \text{Pb} < \text{Fe}$ (D) $\text{Zn} < \text{Fe} < \text{Pb}$

83. The enthalpy of formation, ΔH_f° , equals zero at 25°C for which of the following in their standard states?

- (A) elements (B) compounds
(C) gases (D) solids

84. Liquid water is injected into an oven at 400 K . What are the signs for ΔG , ΔH , and ΔS for the physical transformation that occurs?

	ΔG	ΔH	ΔS
(A)	+	-	-
(B)	+	-	0
(C)	-	+	+
(D)	-	+	0

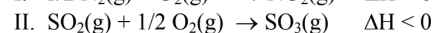
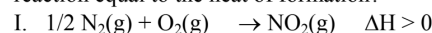
- (A) (A) (B) (B) (C) (C) (D) (D)

85. A sample of $\text{NaOH}(\text{s})$ was added to water in a calorimeter. The temperature was monitored as the NaOH dissolved to give the data below. Determine the heat released during the solution process. (Assume the solution specific heat is $4.18 \text{ J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$)

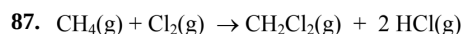
Mass of water	100.00 g
Mass of $\text{NaOH}(\text{s})$	10.00 g
Initial Temperature of water	24.0°C
Final Temperature of solution	48.2°C

- (A) $1.01 \times 10^3 \text{ Joules}$ (B) $2.66 \times 10^3 \text{ Joules}$
(C) $1.01 \times 10^4 \text{ Joules}$ (D) $1.11 \times 10^4 \text{ Joules}$

86. For which of the reactions below is(are) the heat of reaction equal to the heat of formation?



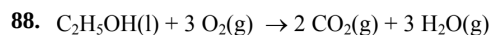
- (A) I. only (B) II. only
(C) Both I. and II. (D) Neither I. nor II.



Use the data in the table below to calculate the standard enthalpy, ΔH° , for the reaction above.

Substance	$\text{CH}_4(\text{g})$	$\text{CH}_2\text{Cl}_2(\text{g})$	$\text{HCl}(\text{g})$
$\Delta H_f^\circ, \text{kJ}\cdot\text{mol}^{-1}$	-74.6	-95.4	-92.3

- (A) $-205 \text{ kJ}\cdot\text{mol}^{-1}$ (B) $-113 \text{ kJ}\cdot\text{mol}^{-1}$
(C) $113 \text{ kJ}\cdot\text{mol}^{-1}$ (D) $205 \text{ kJ}\cdot\text{mol}^{-1}$

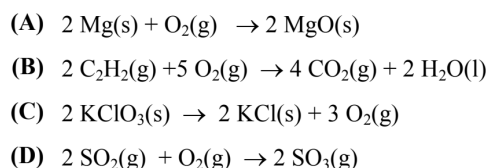


During an experiment 10.00 g of ethanol is completely burned in air to release $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$ as shown in the equation above. During the combustion, 296.6 kJ of heat energy is released. What is the molar enthalpy of combustion, $\Delta H_{\text{comb}}^\circ$?

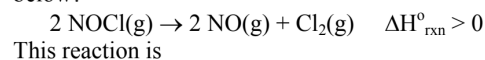
- (A) $-2966 \text{ kJ}\cdot\text{mol}^{-1}$ (B) $-1366 \text{ kJ}\cdot\text{mol}^{-1}$
(C) $-64.36 \text{ kJ}\cdot\text{mol}^{-1}$ (D) $-29.66 \text{ kJ}\cdot\text{mol}^{-1}$

Thermodynamics

89. Which reaction has the greatest positive change in entropy, ΔS ?



90. Which statement is correct for the reaction represented below?



- (A) spontaneous at all temperatures.
 (B) spontaneous only at high temperatures.
 (C) spontaneous only at low temperatures.
 (D) not spontaneous at any temperature.

91. The combustion of 1.00 mole of methane, CH_4 , produces carbon dioxide and water and releases $802 \text{ kJ} \cdot \text{mol}^{-1}$. When 3.00 mol of oxygen react with a stoichiometric quantity of methane, what is ΔH for the reaction?

- (A) $-1.20 \times 10^3 \text{ kJ}$ (B) $-8.02 \times 10^2 \text{ kJ}$
 (C) $8.02 \times 10^2 \text{ kJ}$ (D) $1.20 \times 10^3 \text{ kJ}$

92. What is the final temperature when 20.0 g of water (specific heat capacity $4.184 \text{ J} \cdot \text{g}^{-1} \cdot \text{K}^{-1}$) at 80.0°C is mixed with 30.0 g of water at 20.0°C in an insulated container?

- (A) 32°C (B) 44°C (C) 50°C (D) 56°C

93. $\text{PCl}_5\text{(s)}$ reacts with $\text{H}_2\text{O(l)}$ according to the equation:
 $\text{PCl}_5\text{(s)} + 4\text{H}_2\text{O(l)} \rightarrow \text{H}_3\text{PO}_4\text{(aq)} + 5\text{HCl(aq)}$
 What is ΔH° for this reaction in $\text{kJ} \cdot \text{mol}^{-1}$?

Substance	$\text{PCl}_5\text{(s)}$	$\text{H}_2\text{O(l)}$	$\text{H}_3\text{PO}_4\text{(aq)}$	HCl(aq)
$\Delta H_f^\circ, \text{ kJ/mol}$	-443.5	-285.8	-1284.4	-167.1

- (A) -722.2 (B) -533.2 (C) 533.2 (D) 722.2

94. Which is a statement of the Second Law of Thermodynamics?

- (A) The energy of the universe is conserved.
 (B) The energy of the universe is decreasing.
 (C) The entropy of the universe is conserved.
 (D) The entropy of the universe is increasing.

95. For which signs of ΔH and ΔS will a reaction always be spontaneous?

- (A) $\Delta H +, \Delta S +$ (B) $\Delta H +, \Delta S -$
 (C) $\Delta H -, \Delta S -$ (D) $\Delta H -, \Delta S +$

96. What is correct about the signs and magnitudes of the free energy, ΔG° , and the equilibrium constant, K , for a thermodynamically spontaneous reaction under standard conditions?

- (A) $\Delta G^\circ < 0, K < 0$ (B) $\Delta G^\circ = 0, K > 0$
 (C) $\Delta G^\circ < 0, K = 0$ (D) $\Delta G^\circ < 0, K > 0$

97. The energies of the bonds broken in a certain reaction are greater than the energies of the bonds formed. Which one of the following statements about this reaction must be true?

- (A) The reaction is endothermic.
 (B) The reaction is exothermic.
 (C) The reaction is spontaneous.
 (D) The reaction is non-spontaneous.

98. How much heat is required to convert 5.0 g of ice at -10.0°C to liquid water at 15.0°C ? (Assume heat capacities are independent of temperature.)

Enthalpy of fusion	$6.00 \text{ kJ} \cdot \text{mol}^{-1}$
Specific heat capacity of ice	$37.8 \text{ J} \cdot \text{mol}^{-1} \cdot ^\circ \text{C}^{-1}$
Specific heat capacity of water	$76.0 \text{ J} \cdot \text{mol}^{-1} \cdot ^\circ \text{C}^{-1}$

- (A) $4.2 \times 10^2 \text{ J}$ (B) $2.1 \times 10^3 \text{ J}$
 (C) $9.3 \times 10^3 \text{ J}$ (D) $3.8 \times 10^4 \text{ J}$

99. What is the standard enthalpy of formation of ethylene, C_2H_4 , if its standard enthalpy of combustion is $-1411 \text{ kJ} \cdot \text{mol}^{-1}$?

substance	$\Delta H_f^\circ, \text{ kJ} \cdot \text{mol}^{-1}$
$\text{CO}_2\text{(g)}$	-394
$\text{H}_2\text{O(l)}$	-286

- (A) $1411 \text{ kJ} \cdot \text{mol}^{-1}$ (B) $51 \text{ kJ} \cdot \text{mol}^{-1}$
 (C) $-337 \text{ kJ} \cdot \text{mol}^{-1}$ (D) $-445 \text{ kJ} \cdot \text{mol}^{-1}$

100. At the triple point of water how do the entropies of solid, liquid, and gas compare?

- (A) $S^\circ_{\text{solid}} < S^\circ_{\text{liquid}} < S^\circ_{\text{gas}}$ (B) $S^\circ_{\text{solid}} = S^\circ_{\text{liquid}} < S^\circ_{\text{gas}}$
 (C) $S^\circ_{\text{gas}} < S^\circ_{\text{solid}} < S^\circ_{\text{liquid}}$ (D) $S^\circ_{\text{solid}} = S^\circ_{\text{liquid}} = S^\circ_{\text{gas}}$

101. "A perfect crystalline substance has an entropy of zero at absolute zero" is a statement of

- (A) Hess's Law.
 (B) The First Law of Thermodynamics.
 (C) The Second Law of Thermodynamics.
 (D) The Third Law of Thermodynamics.

Thermodynamics

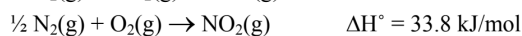
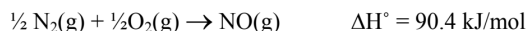
102. Which one of the following sets of conditions would result in a reaction that is spontaneous at high temperatures but non-spontaneous at low temperatures?

(A) $\Delta H > 0$, $\Delta S > 0$ (B) $\Delta H > 0$, $\Delta S < 0$
 (C) $\Delta H < 0$, $\Delta S < 0$ (D) $\Delta H < 0$, $\Delta S > 0$

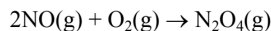
103. Which process is exothermic?

(A) condensation (B) fusion
 (C) sublimation (D) vaporization

104. Use the thermodynamic information:



to calculate ΔH° in kJ/mol for the reaction:



(A) -171.2 (B) -114.6 (C) 114.6 (D) 171.2

105. Determine the enthalpy change for the reaction of 5.00 g of Fe_2O_3 with aluminum metal according to the equation $\text{Fe}_2\text{O}_3(\text{s}) + 2\text{Al}(\text{s}) \rightarrow \text{Al}_2\text{O}_3(\text{s}) + 2\text{Fe}(\text{l})$

Substance	ΔH_f° kJ/mol
$\text{Fe}_2\text{O}_3(\text{s})$	-825.5
$\text{Al}_2\text{O}_3(\text{s})$	-1675.7
$\text{Fe}(\text{l})$	12.4

(A) -25.8 kJ (B) -26.2 kJ
 (C) -52.4 kJ (D) -77.9 kJ

106. Which reaction has the most positive entropy change under standard conditions?

(A) $\text{H}_2\text{O}(\text{g}) + \text{CO}(\text{g}) \rightarrow \text{H}_2(\text{g}) + \text{CO}_2(\text{g})$
 (B) $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
 (C) $\text{NH}_3(\text{g}) \rightarrow \text{NH}_3(\text{aq})$
 (D) $\text{C}_8\text{H}_{18}(\text{l}) \rightarrow \text{C}_8\text{H}_{18}(\text{s})$

107. What are the signs of ΔH and ΔS for a reaction that is spontaneous only at low temperatures?

(A) ΔH is positive, ΔS is positive
 (B) ΔH is positive, ΔS is negative
 (C) ΔH is negative, ΔS is negative
 (D) ΔH is negative, ΔS is positive

108. Which substance has a non-zero standard free energy of formation?

(A) $\text{Pb}(\text{s})$ (B) $\text{Hg}(\text{l})$ (C) $\text{Cl}_2(\text{g})$ (D) $\text{O}_3(\text{g})$

109. A 22.0 g piece of metal is heated to 100.0°C and placed in 75.0 g H_2O at 25.0°C . If the final temperature of the metal and water is 27.8°C , what is the specific heat capacity of the metal in $\text{J}\cdot\text{g}^{-1}\cdot^\circ\text{C}^{-1}$? (Assume no heat is lost/gained by the surroundings.)

$C_p / \text{J}\cdot\text{g}^{-1}\cdot^\circ\text{C}^{-1}$
H_2O 4.18

(A) 0.038 (B) 0.16 (C) 0.55 (D) 5.0

110. Which change(s) is(are) accompanied by an increase in entropy of the system?

I. conversion of $\text{O}_2(\text{g})$ to $\text{O}_3(\text{g})$
 II. freezing of water
 III. sublimation of iodine

(A) I only (B) III only
 (C) I and II only (D) II and III only

111. $\text{NO}(\text{g}) \rightarrow \frac{1}{2} \text{N}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \quad \Delta H_1^\circ$
 $2\text{NO}(\text{g}) \rightarrow \text{N}_2\text{O}(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \quad \Delta H_2^\circ$

Which relationship is correct?

(A) $\Delta H_1^\circ = \Delta H_2^\circ$
 (B) ΔH_f° for $\text{NO}(\text{g}) = \Delta H_1^\circ$
 (C) ΔH_f° for $\text{N}_2\text{O}(\text{g}) = \Delta H_2^\circ$
 (D) ΔH_f° for $\text{N}_2\text{O}(\text{g}) = \Delta H_2^\circ - 2\Delta H_1^\circ$

112. When 2.74 g of $\text{Ba}(\text{s})$ reacts with $\text{O}_2(\text{g})$ at 298 K and 1 atm to form $\text{BaO}(\text{s})$, 11,100 J of heat is released. What is ΔH_f° for $\text{BaO}(\text{s})$ in $\text{kJ}\cdot\text{mol}^{-1}$?

(A) 556 (B) 221 (C) -221 (D) -556

113. A reaction has $\Delta H^\circ > 0$ and $\Delta G^\circ > 0$ at 25°C . This reaction

(A) is at equilibrium at 25°C .
 (B) could not be spontaneous under standard conditions at any temperature.
 (C) could be spontaneous under standard conditions at temperatures above 25°C .
 (D) could be spontaneous under standard conditions at temperatures below 25°C .

114. An ionic compound has a solubility of $1 \text{ mol}\cdot\text{L}^{-1}$ in water at 25°C and its solubility increases as the temperature is raised. What are the signs of ΔH° and ΔS° for the dissolving process?

	ΔH°	ΔS°
(A)	+	+
(B)	+	-
(C)	-	+
(D)	-	-

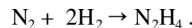
Thermodynamics

115. Which statement is always true for an exothermic reaction?
- (A) The enthalpy change is negative.
 (B) The entropy change is negative.
 (C) The reaction absorbs heat from the surroundings.
 (D) The reaction is spontaneous.
116. For which reaction is $\Delta H_{\text{rxn}}^\circ$ equal to ΔH_f° for $\text{CuSO}_4(\text{s})$?
- (A) $\text{Cu}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{CuSO}_4(\text{s})$
 (B) $\text{CuO}(\text{s}) + \text{SO}_3(\text{g}) \rightarrow \text{CuSO}_4(\text{s})$
 (C) $\text{CuS}(\text{s}) + 2\text{O}_2(\text{g}) \rightarrow \text{CuSO}_4(\text{s})$
 (D) $\text{Cu}(\text{s}) + \text{S}(\text{s}) + 2\text{O}_2(\text{g}) \rightarrow \text{CuSO}_4(\text{s})$
117. What is the value of ΔH° (in kJ) for this reaction?
- | | $\Delta H_f^\circ / \text{kJ}\cdot\text{mol}^{-1}$ |
|-----------------------|--|
| CuO | -156.1 |
| Cu_2O | -170.7 |
- $2\text{CuO}(\text{s}) \rightarrow \text{Cu}_2\text{O}(\text{s}) + \frac{1}{2}\text{O}_2(\text{g})$
- (A) 141.5 (B) 14.6 (C) -14.6 (D) -141.5
118. When the substances $\text{Na}(\text{s})$, $\text{Br}_2(\text{g})$, $\text{Br}_2(\text{l})$, $\text{NaBr}(\text{s})$ are arranged in order of increasing molar entropy at 25°C , which order is correct?
- (A) $\text{Br}_2(\text{g})$, $\text{Br}_2(\text{l})$, $\text{Na}(\text{s})$, $\text{NaBr}(\text{s})$
 (B) $\text{Na}(\text{s})$, $\text{Br}_2(\text{g})$, $\text{Br}_2(\text{l})$, $\text{NaBr}(\text{s})$
 (C) $\text{Na}(\text{s})$, $\text{NaBr}(\text{s})$, $\text{Br}_2(\text{l})$, $\text{Br}_2(\text{g})$
 (D) $\text{NaBr}(\text{s})$, $\text{Br}_2(\text{g})$, $\text{Br}_2(\text{l})$, $\text{Na}(\text{s})$
119. The reaction; $4\text{Ag}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{Ag}_2\text{O}(\text{s})$, is exothermic. Which statement about the reaction is correct?
- (A) It is spontaneous at all temperatures.
 (B) It is spontaneous only at low temperatures.
 (C) It is spontaneous only at high temperatures.
 (D) It is non-spontaneous at all temperatures.
120. For a reaction at 25°C , $\Delta G^\circ = -33.3 \text{ kJ}$ and $\Delta S^\circ = -198 \text{ J}\cdot\text{K}^{-1}$. What is the value of ΔH° ?
- (A) -92.3 kJ (B) -38.3 kJ
 (C) -28.8 kJ (D) 25.7 kJ
121. The standard enthalpy of formation for $\text{NH}_3(\text{g})$ is $-46.1 \text{ kJ}\cdot\text{mol}^{-1}$. Calculate ΔH° for the reaction:
- $2\text{NH}_3(\text{g}) \rightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$
- (A) -92.2 kJ (B) -46.1 kJ
 (C) 46.1 kJ (D) 92.2 kJ
122. What is the specific heat capacity of mercury (in $\text{J}\cdot\text{g}^{-1}\cdot^\circ\text{C}^{-1}$) if a 25.0 g sample requires 19.3 J to raise its temperature from 24.5°C to 30.0°C ?
- (A) 0.026 (B) 0.032 (C) 0.14 (D) 7.1
123. Which are exothermic processes?
- | |
|--|
| I. combustion of ethane |
| II. dehydration of barium chloride dihydrate |
- (A) I only (B) II only
 (C) both I and II (D) neither I nor II
124. Which has the highest standard molar entropy?
- (A) $\text{O}_2(\text{g})$ (B) $\text{SO}_2(\text{g})$ (C) $\text{H}_2\text{O}(\text{l})$ (D) $\text{PbO}_2(\text{s})$
125. Which is always true for a specific system during a spontaneous reaction?
- (A) $\Delta H < 0$ (B) $\Delta H \geq 0$ (C) $\Delta G < 0$ (D) $\Delta S > 0$
126. Calculate the change in enthalpy (in kJ per mole of CO_2) for the decomposition of sodium hydrogen carbonate from the standard enthalpies of formation:
- | | $\Delta H_f^\circ (\text{kJ}\cdot\text{mol}^{-1})$ |
|------------------------------------|--|
| $\text{NaHCO}_3(\text{s})$ | -947.7 |
| $\text{Na}_2\text{CO}_3(\text{s})$ | -1130.9 |
| $\text{H}_2\text{O}(\text{g})$ | -241.8 |
| $\text{CO}_2(\text{g})$ | -393.5 |
- $2 \text{NaHCO}_3(\text{s}) \rightarrow \text{Na}_2\text{CO}_3(\text{s}) + \text{H}_2\text{O}(\text{g}) + \text{CO}_2(\text{g})$
- (A) 129.2 (B) -818.5 (C) -1766.2 (D) -3661.6
127. A chemical reaction is carried out twice with the same quantity of reactants to form the same products but the pressure is different for the two experiments. Which does not change?
- (A) K_p (B) heat released
 (C) $\Delta T_{\text{surroundings}}$ (D) work done
128. Which of these conversions has a positive ΔS° ?
- | |
|---|
| I. combustion of charcoal |
| II. condensation of $\text{Br}_2(\text{g})$ |
| III. precipitation of $\text{AgCl}(\text{s})$ |
- (A) I only (B) II only
 (C) III only (D) II and III only
129. Given these reactions:
- $\text{A} \rightarrow 2\text{B}$ $\Delta H = 40 \text{ kJ}$
 $\text{B} \rightarrow \text{C}$ $\Delta H = -50 \text{ kJ}$
 $2\text{C} \rightarrow \text{D}$ $\Delta H = -20 \text{ kJ}$
- Calculate ΔH for the reaction; $\text{D} + \text{A} \rightarrow 4\text{C}$.
- (A) -100 kJ (B) -60 kJ
 (C) -40 kJ (D) 100 kJ

Thermodynamics

130. Hydrazine, N_2H_4 , contains a N-N single bond and 4 N-H bonds. Use bond energies to calculate ΔH in kJ for the reaction;

Bond Energies ($\text{kJ}\cdot\text{mol}^{-1}$)	
H-H	436
N-H	389
N-N	159
N=N	418
N $\equiv$ N	941



- (A) -425 kJ (B) -98 kJ (C) 98 kJ (D) 425 kJ

An ice cube of unknown mass at 0°C is added to 265 g of H_2O at 25.00°C in a

Properties of Water	
C_p	$4.18 \text{ J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$
ΔH_{fusion}	$333 \text{ J}\cdot\text{g}^{-1}$

calorimeter. If the final temperature of the resulting H_2O is 21.70°C , what is the mass of the ice cube?

- (A) 2.47 g (B) 8.63 g (C) 10.3 g (D) 11.0 g

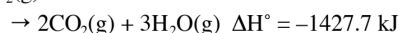
132. Which reaction is spontaneous at all temperatures at standard pressure and concentration?

- (A) exothermic reaction with a decrease in entropy
(B) exothermic reaction with an increase in entropy
(C) endothermic reaction with a decrease in entropy
(D) endothermic reaction with an increase in entropy

133. Which equation represents the reaction for the standard enthalpy of formation, ΔH_f° , for $\text{B}_5\text{H}_9(\text{g})$ at 298 K and 1 atm?

- (A) $5\text{B}(\text{s}) + 9\text{H}(\text{g}) \rightarrow \text{B}_5\text{H}_9(\text{g})$
(B) $2\text{B}(\text{s}) + 3\text{BH}_3(\text{g}) \rightarrow \text{B}_5\text{H}_9(\text{g})$
(C) $\frac{5}{2}\text{B}_2(\text{g}) + \frac{9}{2}\text{H}_2(\text{g}) \rightarrow \text{B}_5\text{H}_9(\text{g})$
(D) $5\text{B}(\text{s}) + \frac{9}{2}\text{H}_2(\text{g}) \rightarrow \text{B}_5\text{H}_9(\text{g})$

134. $\text{C}_2\text{H}_6(\text{g}) + \frac{7}{2}\text{O}_2(\text{g})$



If the enthalpy of vaporization for $\text{H}_2\text{O}(\text{l})$ is 44.0 kJ/mol , what is ΔH° for this reaction if $\text{H}_2\text{O}(\text{l})$ is formed instead of $\text{H}_2\text{O}(\text{g})$?

- (A) -1295.7 kJ (B) -1383.7 kJ
(C) -1471.7 kJ (D) -1559.7 kJ

135. A gold ring that weighs 3.81 g is heated to 84.0°C and placed in 50.0 g of H_2O at 22.1°C . What is the final temperature?

Specific Heat Capacity ($\text{J}\cdot\text{g}^{-1}\cdot^\circ\text{C}^{-1}$)	
Au	0.129
H_2O	4.18

- (A) 22.2°C (B) 24.0°C
(C) 26.5°C (D) 53.1°C

136. Calculate the change in enthalpy, ΔH , for the combustion of 11.2 L of hydrogen gas, measured at 0°C and 1 atm pressure, to form $\text{H}_2\text{O}(\text{g})$.

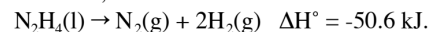
ΔH_f° ($\text{kJ}\cdot\text{mol}^{-1}$)	
$\text{H}_2\text{O}(\text{g})$	-241.8

- (A) -60.5 kJ (B) -121 kJ
(C) -484 kJ (D) -2710 kJ

137. Which reaction proceeds with the greatest increase in entropy?

- (A) $\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}_2(\text{l})$
(B) $\text{Br}_2(\text{l}) + \text{F}_2(\text{g}) \rightarrow 2\text{BrF}(\text{g})$
(C) $\text{Cu}^{2+}(\text{aq}) + \text{Zn}(\text{s}) \rightarrow \text{Cu}(\text{s}) + \text{Zn}^{2+}(\text{aq})$
(D) $4\text{NH}_3(\text{g}) + 7\text{O}_2(\text{g}) \rightarrow 4\text{NO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$

138. For the reaction,



This reaction is

- (A) spontaneous at all temperatures.
(B) non-spontaneous at all temperatures.
(C) spontaneous only at low temperatures.
(D) spontaneous only at high temperatures.

139. The enthalpy change of which reaction corresponds to ΔH_f° for $\text{Na}_2\text{CO}_3(\text{s})$ at 298 K?

- (A) $2\text{Na}(\text{s}) + \text{C}(\text{s}) + \frac{3}{2}\text{O}_2(\text{g}) \rightarrow \text{Na}_2\text{CO}_3(\text{s})$
(B) $\text{Na}_2\text{O}(\text{s}) + \text{CO}_2(\text{g}) \rightarrow \text{Na}_2\text{CO}_3(\text{s})$
(C) $2\text{Na}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{Na}_2\text{CO}_3(\text{s})$
(D) $2\text{Na}^+(\text{aq}) + 2\text{OH}^-(\text{aq}) + \text{CO}_2(\text{aq}) \rightarrow \text{Na}_2\text{CO}_3(\text{s}) + \text{H}_2\text{O}$

140. Which applies to any endothermic reaction?

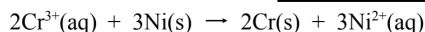
- (A) $\Delta H < 0$ (B) $\Delta H > 0$ (C) $\Delta G < 0$ (D) $\Delta G > 0$

141. When a bomb calorimeter is used to determine the heat of reaction, which property of the system under investigation is most likely to remain constant?

- (A) number of molecules (B) pressure
(C) temperature (D) volume

142. For the reaction shown, which is closest to the value of ΔH ?

ΔH_f° ($\text{kJ}\cdot\text{mol}^{-1}$)	
$\text{Cr}^{3+}(\text{aq})$	-143
$\text{Ni}^{2+}(\text{aq})$	-54



- (A) 124 kJ (B) 89 kJ (C) -89 kJ (D) -124 kJ

Thermodynamics

143. An ice cube at 0.00 °C is placed in 200. g of distilled water at 25.00 °C. The final temperature after the ice is completely melted is 5.00 °C. What is the mass of the ice cube? ($\Delta H_{\text{fus}} = 340. \text{ J}\cdot\text{g}^{-1}$, $C_p = 4.18 \text{ J}\cdot\text{g}^{-1}\cdot^\circ\text{C}^{-1}$)

(A) 23.6 g (B) 46.3 g (C) 50.0 g (D) 800. g

144. Which reaction occurs with the greatest increase in entropy?

(A) $2\text{H}_2\text{O}(l) \rightarrow 2\text{H}_2(g) + \text{O}_2(g)$
 (B) $2\text{NO}(g) \rightarrow \text{N}_2(g) + \text{O}_2(g)$
 (C) $\text{C}(s) + \text{O}_2(g) \rightarrow \text{CO}_2(g)$
 (D) $\text{Br}_2(g) + \text{Cl}_2(g) \rightarrow 2\text{BrCl}(g)$

145. Calculate the amount of energy released when 0.100 mol of diborane, B_2H_6 , reacts with oxygen to produce solid B_2O_3 and steam.

$\Delta H_f^\circ, (\text{kJ}\cdot\text{mol}^{-1})$	
$\text{B}_2\text{H}_6(g)$	35
$\text{B}_2\text{O}_3(s)$	-1272
$\text{H}_2\text{O}(l)$	-285
$\text{H}_2\text{O}(g)$	-241

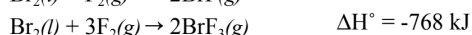
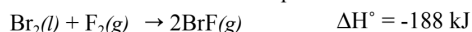
(A) 203 kJ (B) 216 kJ (C) 330 kJ (D) 343 kJ

146. How much heat is required to raise the temperature of 100. g of Fe_2O_3 from 5.0 °C to 25.0 °C?

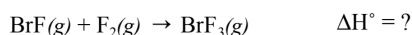
Specific heat, $\text{J}\cdot\text{g}^{-1}\cdot^\circ\text{C}^{-1}$	
Fe_2O_3	0.634

(A) 1.58 kJ (B) 1.27 kJ
 (C) 0.845 kJ (D) 0.0634 kJ

147. Given the thermochemical equations:

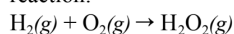


determine ΔH° for the reaction



(A) -956 kJ (B) -580 kJ
 (C) -478 kJ (D) -290 kJ

148. Use bond energies to calculate ΔH° for the reaction:



Bond Energy, $\text{kJ}\cdot\text{mol}^{-1}$	
H-H	432
H-O	459
O-O	207
O=O	494

(A) -521 kJ (B) -486 kJ
 (C) -199 kJ (D) 199 kJ

149. Which reaction occurs with a decrease in entropy?

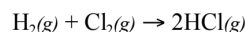
(A) $\text{N}_2(g) + \text{O}_2(g) \rightarrow 2\text{NO}(g)$
 (B) $\text{N}_2\text{O}_4(g) \rightarrow 2\text{NO}_2(g)$
 (C) $2\text{CO}(g) \rightarrow \text{C}(s) + \text{CO}_2(g)$
 (D) $2\text{HCl}(aq) + \text{Ag}_2\text{CO}_3(s) \rightarrow 2\text{AgCl}(s) + \text{CO}_2(g) + \text{H}_2\text{O}(l)$

150. Calculate the amount of energy necessary to heat a 2.5 g ice cube from 0 °C to 23 °C.

Values for H_2O	
C_p	$4.18 \text{ J}\cdot\text{g}^{-1}\cdot^\circ\text{C}^{-1}$
ΔH_{fusion}	$3.4 \times 10^2 \text{ J}\cdot\text{g}^{-1}$

(A) 240 J (B) 850 J
 (C) 1100 J (D) 3700 J

Estimate ΔH for this reaction.



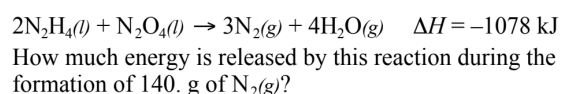
Bond Energies, $\text{kJ}\cdot\text{mol}^{-1}$	
H-H	436
Cl-Cl	243
H-Cl	431

(A) 1110 kJ (B) 248 kJ
 (C) -183 kJ (D) -248 kJ

152. Which reaction occurs with an increase in entropy?

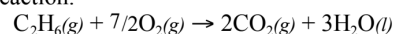
(A) $2\text{C}(s) + \text{O}_2(g) \rightarrow 2\text{CO}(g)$
 (B) $2\text{H}_2\text{S}(g) + \text{SO}_2(g) \rightarrow 3\text{S}(s) + 2\text{H}_2\text{O}(g)$
 (C) $4\text{Fe}(s) + 3\text{O}_2(g) \rightarrow 2\text{Fe}_2\text{O}_3(s)$
 (D) $\text{CO}(g) + 2\text{H}_2(g) \rightarrow \text{CH}_3\text{OH}(l)$

153. Consider this reaction.



(A) 1078 kJ (B) 1797 kJ
 (C) 3234 kJ (D) 5390 kJ

154. Use the information in the table to calculate the enthalpy of this reaction.



Reaction	$\Delta H_f^\circ, \text{kJ}\cdot\text{mol}^{-1}$
$2\text{C}(s) + 3\text{H}_2(g) \rightarrow \text{C}_2\text{H}_6(g)$	-84.7
$2\text{C}(s) + \text{O}_2(g) \rightarrow \text{CO}_2(g)$	-393.5
$\text{H}_2(g) + 1/2\text{O}_2(g) \rightarrow \text{H}_2\text{O}(l)$	-285.8

(A) -764 kJ (B) -1560 kJ
 (C) -1664 kJ (D) -3120 kJ

Thermodynamics

155. For the reaction $\text{PCl}_3(g) + \text{Cl}_2(g) \rightarrow \text{PCl}_5(g)$, $\Delta H^\circ = -86 \text{ kJ}$. Under what temperatures is this reaction expected to be spontaneous?

(A) no temperatures (B) high temperatures only
(C) all temperatures (D) low temperatures only

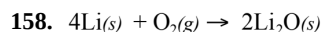
156. For the formation of one mole of each of these gases from their elements, which reaction is most endothermic?

(A) CO ($\Delta H_f^\circ = -110.5 \text{ kJ}\cdot\text{mol}^{-1}$)
(B) NO_2 ($\Delta H_f^\circ = +33.9 \text{ kJ}\cdot\text{mol}^{-1}$)
(C) O_3 ($\Delta H_f^\circ = +142.2 \text{ kJ}\cdot\text{mol}^{-1}$)
(D) SO_2 ($\Delta H_f^\circ = -300.4 \text{ kJ}\cdot\text{mol}^{-1}$)

157. The specific heats of several metals are given in the table. If the same number of Joules were applied to the same mass of each metal, which metal would show the greatest temperature change?

Specific Heat, $\text{J}\cdot\text{g}^{-1}\cdot^\circ\text{C}^{-1}$	
Al	0.900
Au	0.129
Cu	0.385
Hg	0.139

(A) Al (B) Au (C) Cu (D) Hg



At 25°C , ΔH° for this reaction is -598.8 kJ per mole of $\text{Li}_2\text{O}(s)$ formed. What mass of Li should be reacted with excess $\text{O}_2(g)$ in order to release $150. \text{ kJ}$?

(A) 0.874 g (B) 1.74 g
(C) 3.48 g (D) 6.98 g

159. When these substances are arranged in order of increasing S° values at 25°C , what is the correct order?

(A) $\text{Na}(s)$, $\text{Cl}_2(g)$, $\text{NaCl}(s)$ (B) $\text{NaCl}(s)$, $\text{Cl}_2(g)$, $\text{Na}(s)$
(C) $\text{Cl}_2(g)$, $\text{NaCl}(s)$, $\text{Na}(s)$ (D) $\text{Na}(s)$, $\text{NaCl}(s)$, $\text{Cl}_2(g)$

160. The ΔH° and ΔS° values for a particular reaction are -60.0 kJ and $-0.200 \text{ kJ}\cdot\text{K}^{-1}$ respectively. Under what conditions is this reaction spontaneous?

(A) all conditions (B) $T < 300 \text{ K}$
(C) $T = 300 \text{ K}$ (D) $T > 300 \text{ K}$

161. For which of these is ΔH_f° not equal to zero?

(A) $\text{Br}_2(l)$ (B) $\text{Fe}(s)$ (C) $\text{I}_2(s)$ (D) $\text{O}_3(g)$

162. The enthalpy change for which reaction represents the standard enthalpy of formation for hydrogen cyanide, HCN ?

(A) $\text{H}(g) + \text{C}(\text{graphite}) + \text{N}(g) \rightarrow \text{HCN}(g)$
(B) $\frac{1}{2}\text{H}_2(g) + \text{C}(\text{graphite}) + \frac{1}{2}\text{N}_2(g) \rightarrow \text{HCN}(g)$
(C) $\text{HCN}(g) \rightarrow \frac{1}{2}\text{H}_2(g) + \text{C}(\text{graphite}) + \frac{1}{2}\text{N}_2(g)$
(D) $\text{H}_2(g) + 2\text{C}(\text{graphite}) + \text{N}_2(g) \rightarrow 2\text{HCN}(g)$

163. What is the standard enthalpy of formation of $\text{MgO}(s)$ if 300.9 kJ is evolved when 20.15 g of $\text{MgO}(s)$ is formed by the combustion of magnesium under standard conditions?

(A) $-601.8 \text{ kJ}\cdot\text{mol}^{-1}$ (B) $-300.9 \text{ kJ}\cdot\text{mol}^{-1}$
(C) $+300.9 \text{ kJ}\cdot\text{mol}^{-1}$ (D) $+601.8 \text{ kJ}\cdot\text{mol}^{-1}$

164. Which combination of solutions of HCl and NaOH would produce the largest ΔT ?

(A) 50 mL of 1 M HCl with 50 mL of 1 M NaOH
(B) 50 mL of 2 M HCl with 50 mL of 2 M NaOH
(C) 100 mL of 1 M HCl with 50 mL of 2 M NaOH
(D) 100 mL of 1 M HCl with 100 mL of 1 M NaOH

165. Which change occurs with the largest increase in entropy at 25°C ?

(A) $\text{Br}_2(l) \rightarrow \text{Br}_2(g)$
(B) $\text{C}(\text{graphite}) \rightarrow \text{C}(\text{diamond})$
(C) $\text{H}_2\text{O}(s) \rightarrow \text{H}_2\text{O}(l)$
(D) $\text{HCl}(g) + \text{H}_2\text{O}(l) \rightarrow \text{H}_3\text{O}^+(aq) + \text{Cl}^-(aq)$

166. What are the signs of ΔH° and ΔS° for a reaction that is spontaneous at all temperatures?

	ΔH°	ΔS°
(A)	+	+
(B)	+	-
(C)	-	+
(D)	-	-